

The Cost of Reliability in Optimal Magnetization Switching: A Longitudinal-Field Front for Van der Waals Magnets

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Abstract

Optimal control theory delivers magnetic-field pulses that reverse a magnetic moment for the least dissipated energy, and recent work shows these pulses reach picosecond, energy-efficient switching in two-dimensional van der Waals magnets. Because the optimal pulse is always perpendicular to the moment, it does nothing to damp the perturbations that thermal fluctuations excite, and the switching becomes unreliable exactly where it matters. A longitudinal field parallel to the moment removes that instability and is invisible to the leading-order dynamics, so it is often described as free. It is not free: it costs the square of its amplitude integrated over the pulse. We compute, for the first time, the cost-reliability front of that longitudinal field. Using an open-source engine that solves optimal control over the Landau-Lifshitz-Gilbert equation and a thermostat validated against the Boltzmann distribution, we reproduce the published stabilization physics for CrSBr, the half-hyperbolic bare path and the counterintuitive success dip near one anisotropy field, and we add the missing axis: the added switching cost, which grows quadratically with the field. The price is not small: a field of one anisotropy field already costs 2.5 times the bare optimal switching cost, and two anisotropy fields cost ten times it. Nor is the benefit universal. At a thermal-stability factor of twenty the bare optimal pulse already reverses every one of 400 copies, so the field buys nothing and costs a great deal; the reliability gain appears only below a stability factor of about ten, where it lifts the success rate from 0.81 to 0.96 at a factor of two. The front makes the reliability-versus-energy trade explicit and quantitative, and it is reproducible from committed artifacts and a live instance.

1 Introduction

The energy cost of writing a magnetic bit has become a design constraint as data-centric computing grows. Conventional field-driven reversal is slow and expensive, which is why current-driven spin-transfer and spin-orbit torque dominate modern magnetic memory. Optimal control theory (OCT) reopens the field-driven route: by shaping the applied field in time one can reverse the moment in the picosecond range at energies up to two orders of magnitude below conventional field protocols, and in two-dimensional van der Waals magnets this becomes competitive with current-based approaches [1]. The formalism rests on inverting the equation of motion so the applied field is a function of the trajectory, which turns a constrained optimization into an unconstrained one over the reversal path [2].

A feature of the optimal pulse is that it is always perpendicular to the moment [2]. That perpendicularity is efficient, but it means the pulse exerts no restoring torque on the transverse perturbations that thermal noise continuously excites. Linearizing the dynamics about the optimal control path shows that the perturbation growth is governed by the eigenvalues of the energy Hessian, and that a large part of a bare reversal is dynamically unstable (hyperbolic), which is the primary cause of the pulse and the moment losing phase lock [3]. A constant longitudinal field, parallel to the instantaneous moment direction, shifts those eigenvalues and removes the hyperbolic domain, which in that study drives the switching success rate to unity; how much reliability it adds depends on how often the bare pulse fails in the first place, which Section 3 measures.

Because the optimal pulse is perpendicular, the longitudinal field does no work at leading order and is frequently treated as a cost-free stabilizer. This is misleading. The switching cost is the time integral of the squared total field, so a longitudinal component of amplitude B_r adds exactly $\int_0^T B_r^2 dt$ to the cost. The trade between reliability and energy is therefore real, and to our knowledge it has not been quantified. We quantify it here.

2 Model and methods

2.1 Dynamics and cost

We consider a single-domain moment with easy-axis anisotropy energy K , magnetic moment μ , gyromagnetic ratio γ and Gilbert damping α . The Landau-Lifshitz-Gilbert (LLG) equation in Gilbert form is

$$(1 + \alpha^2) \dot{\vec{s}} = -\gamma \vec{s} \times (\vec{b}_i + \vec{b}) - \alpha \gamma \vec{s} \times [\vec{s} \times (\vec{b}_i + \vec{b})], \quad (1)$$

with $\vec{b}_i = -\mu^{-1} \partial E / \partial \vec{s}$ the internal field and $\vec{b}$ the applied field. The switching cost, proportional to the Joule heating of the circuit that generates the field, is

$$\Phi = \int_0^T |\vec{b}(t)|^2 dt, \quad (2)$$

in units of T^2 s. Equation (2) is not an energy: converting it to joules requires a circuit constant with units of joules per T^2 s, which we do not assume here, so all costs are reported in their model-free units.

2.2 The optimal pulse and its stability

For a uniaxial magnet, $E = -Ks_z^2$, the optimal control path and its perpendicular pulse are known in closed form as Jacobi elliptic functions [2]. Linearizing the LLG about that path gives a two-component transverse perturbation whose growth is set by

$$w_1 = B_r + \frac{K}{\mu} \cos 2\theta, \quad w_2 = B_r + \frac{K}{\mu} \cos^2 \theta, \quad (3)$$

where θ is the polar angle along the path and B_r is the applied-field component parallel to the moment [3]. Perturbations are bounded (elliptic) when $w_1 w_2 > 0$ and divergent (hyperbolic) when $w_1 w_2 \leq 0$. At $B_r = 0$ the reversal is hyperbolic for $\pi/4 < \theta < 3\pi/4$, exactly half the path. Adding $|B_r| > K/\mu$ removes the hyperbolic domain.

2.3 Thermostat and success rate

We integrate an ensemble of stochastic LLG trajectories with a fluctuating thermal field whose strength is fixed by the fluctuation-dissipation theorem, each Cartesian component drawn per step from a Gaussian of variance $2\alpha k_B T / (\gamma \mu dt)$ [4], using a norm-preserving stochastic Heun

scheme. The thermostat is validated against the Boltzmann distribution it must reproduce: near a uniaxial minimum the mean square polar angle is $\langle \theta^2 \rangle = k_B T / K$, and the thermostat recovers it to within ten percent. For each longitudinal field B_r we apply the bare optimal pulse together with a constant B_r along the instantaneous moment direction, run the ensemble at temperature, and record the fraction that ends in the reversed basin (the success rate) and the added cost $B_r^2 T$.

2.4 Material and engine

The material is monolayer CrSBr, whose measured exchange Hamiltonian and triaxial, substrate-tunable anisotropy make it the natural host of the internal-torque mechanisms of this family [5, 6]. Its magnetic parameters are curated from the primary sources with provenance. All computation is done by `spinopt`, an open-source (MIT) engine that solves optimal control over LLG dynamics; the results here are reproducible from committed artifacts.

3 Results

We first compute the cost-reliability front for CrSBr at a thermal stability factor $\Delta = K/(k_B T) = 20$. Table 1 reports the hyperbolic fraction of the path, the Monte-Carlo success rate over 400 copies with its binomial confidence interval, and the added cost, as the longitudinal field is swept in units of the anisotropy field K/μ .

Table 1: The longitudinal-field cost-reliability front for CrSBr at $\Delta = 20$. The bare path ($B_r = 0$) is half-hyperbolic; a field of order the anisotropy field removes the instability; the added cost grows as B_r^2 . Success rates are over 400 stochastic copies.

$B_r/(K/\mu)$	hyperbolic fraction	success rate	added cost (T^2 s)
0.0	0.50	1.000 ± 0.005	0.00
0.5	0.33	0.980 ± 0.014	6.1×10^{-12}
1.0	0.00	0.958 ± 0.020	2.5×10^{-11}
1.5	0.00	0.998 ± 0.005	5.5×10^{-11}
2.0	0.00	1.000 ± 0.005	9.8×10^{-11}
2.5	0.00	1.000 ± 0.005	1.5×10^{-10}

Three features are worth drawing out. First, the hyperbolic fraction falls from one half at $B_r = 0$ to zero once $B_r \geq K/\mu$, exactly as the linearized stability analysis predicts [3]. Second, the success rate shows a shallow, counterintuitive dip around $B_r \approx K/\mu$ before recovering to unity, the same non-monotonicity reported in the stabilization study, which arises because the eigenvalue ratio passes through -1 near the barrier top in that range. Third, and new here, the added cost grows quadratically with the field. Measured against the bare optimal cost at this switching time, 9.7×10^{-12} T^2 s, it is 0.63 times the bare cost at half an anisotropy field, 2.5 times at one, 10.1 times at two and 15.8 times at two and a half.

Read together, the first and third features say something the stability analysis alone does not. At $\Delta = 20$ the bare pulse is dynamically half-hyperbolic and yet reverses every copy: the thermal kicks are too weak, over ten Larmor times, to exploit the instability. The longitudinal field removes an instability that was not costing anything, at a price several times the switching itself.

Where the field pays: the stability-factor dependence

The reliability gain therefore has to be looked for at lower stability. Table 2 repeats the measurement across the stability factor, comparing the bare optimal pulse with the same pulse

plus a longitudinal field of two anisotropy fields, over 600 stochastic copies per point with 900 integration steps.

Table 2: Success rate of the bare optimal pulse and of the same pulse with $B_r = 2 K/\mu$, against the thermal stability factor, for CrSBr at ten Larmor times; 600 copies per point, 95 per cent binomial intervals. The added cost of the longitudinal field is $9.8 \times 10^{-11} \text{ T}^2 \text{ s}$ at every point, 10.1 times the bare optimal cost. The single-site temperatures are sub-kelvin because one CrSBr site carries an anisotropy of 1.7 K in temperature units; device-grade stability comes from the exchange-coupled volume, not from one site.

$\Delta = K/(k_B T)$	temperature (K)	bare pulse	with $B_r = 2 K/\mu$
1	1.74	0.735 ± 0.035	0.860 ± 0.028
2	0.87	0.810 ± 0.031	0.963 ± 0.015
3	0.58	0.870 ± 0.027	0.983 ± 0.010
5	0.35	0.932 ± 0.020	0.995 ± 0.006
10	0.17	0.983 ± 0.010	1.000 ± 0.003
20	0.09	0.998 ± 0.003	1.000 ± 0.003

The longitudinal field raises the success rate at every stability factor, and the gain is large where the bare pulse is fragile: from 0.81 to 0.96 at $\Delta = 2$, a halving and more of the failure rate, and from 0.87 to 0.98 at $\Delta = 3$. Above $\Delta \approx 10$ the bare pulse already succeeds essentially always and the gain falls inside the confidence interval. At $\Delta = 20$ this independent run, with more copies and a finer step, gives 0.998 ± 0.003 for the bare pulse, consistent with the 1.000 ± 0.005 of Table 1.

The consequence for device design is that the longitudinal field is a stabilizer with a steep price, worth paying only where the bare pulse actually fails. Table 1 lets a designer read off the smallest field that removes the dynamical instability; Table 2 tells them whether, at their stability factor, removing it buys any reliability at all. For a single CrSBr site at ten Larmor times the answer is yes below a stability factor of about ten and no above it.

4 Discussion

The longitudinal-field front reframes reliability as a purchasable quantity with a stated price, rather than a binary the optimal pulse either has or lacks. The bare optimal pulse minimizes energy but is dynamically fragile; the stabilized pulse is reliable but costs more; the front is the locus of the efficient trades between them. Because the added cost is separable, $\int B_r^2 dt$, and the bare cost is the closed-form optimal-control cost, the whole front is cheap to compute once the stability eigenvalues, which come from the same Hessian the optimal-control solver already uses, are in hand.

This suggests a deterministic surrogate for the expensive stochastic sweep. The hyperbolicity integral $\int_0^T \max(0, -w_1 w_2) dt$ along a path is a purely deterministic measure of dynamical fragility that requires no ensemble. Whether penalizing it predicts the Monte-Carlo success rate, and therefore whether a single deterministic optimal-control solve can target a reliability without any stochastic simulation, is a well-posed question this front makes answerable and which we leave to future work.

5 Conclusion

We have quantified the cost of reliability in optimal magnetization switching. For monolayer CrSBr, a longitudinal field of order the anisotropy field removes the dynamical instability of the bare optimal pulse, at an added switching cost that grows quadratically with the field and

exceeds the bare cost already at one anisotropy field. Whether that buys reliability depends on the thermal stability: below a stability factor of about ten it lifts the success rate substantially, and above it the bare pulse is already reliable and the field only adds cost. The reliability-versus-energy trade is now explicit and quantitative, computed by an open engine and reproducible from committed artifacts and a live instance at <https://espira.fasl-work.com>.

Data and code availability

The engine `spinoc` is open source (MIT) at https://github.com/fsantibanezleal/CAOS_SpinOCT. The product, the curated material parameter database, the canonical bake and the companion web application are open (MIT) at https://github.com/fsantibanezleal/CAOS_RES_Espira, with a live instance at <https://espira.fasl-work.com>. Every number in Table 1 is reproduced from the committed artifact `data/artifacts/novel.json`, and every number in Table 2 from the committed artifact `data/artifacts/thermal-success-rate.json` (case C07 of the product’s case registry, release 0.07.000).

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